MINI-PROJECT 2

# Real-time Study Room Booking App

Short Technical Report · React Native & Expo


STUDENT
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SUBMISSION DATE
**10/10/2026**

ROLE
**Full-stack Mobile Developer**

CONTRIBUTION
**100%**


## 1. Deliverable links

**GitHub:** github.com/bunwg29/room-booking

**APK:** `builds/VKU-Room-Booking-v1.0.0.apk`

**Install:** EAS Build eabbb550 · **Video:** Not provided.

**Scope note.** All required on-device booking features are implemented. Room data and realtime availability are local demo data; multi-device sync requires a backend that is outside the supplied repository.

## 2. Feature implementation checklist

| # | Required feature | Status | Implementation evidence |
|---|---|---|---|
| 1 | Room discovery + FlatList | **Complete** | 12 typed rooms, responsive 1/2/3-column list, photos, capacity and live status. |
| 2 | Multi-parameter filters | **Complete** | Search; status; buildings A/B/C/V; capacity 2–20; four equipment types. |
| 3 | 7-day time-slot selector | **Complete** | Seven dates and four discrete 2-hour slots. |
| 4 | Conflict prevention | **Complete** | UI disabling plus store-level interval validation. |
| 5 | Booking pass + QR check-in | **Complete** | Unique pass and real scannable QR payload. |

IMPLEMENTATION

## 3. Technical architecture

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| Expo Router screens | React Query + services | Zustand actions | AsyncStorage + notifications |

### Project structure

```
src/
├── app/          routes & tabs
├── components/   cards & QR modal
├── data/         room catalogue
├── hooks/        React Query hooks
├── screens/      feature screens
├── services/     rules & reminders
├── store/        persisted Zustand
└── types/        domain models
```

### State flow

Rooms load through a typed data-source boundary and React Query. A booking action enters the Zustand store, which rechecks conflicts, requests notification permission, persists the booking, then immediately updates every subscribed screen.

Static room occupancy and active local bookings are combined into derived "Available Now / Occupied" UI states.

### Conflict rule

```
sameRoom && sameDate && newStart < existingEnd && newEnd > existingStart
```

The rule is applied both when painting disabled slots and again inside `addBooking`. This second guard prevents stale UI or rapid taps from inserting overlapping reservations.

### Booking lifecycle

**CONFIRMED**
Persist booking, generate pass, schedule reminder.

**CHECKED_IN / CANCELLED**
Persist transition and cancel scheduled reminder.

## 5. Technical challenges & resolutions

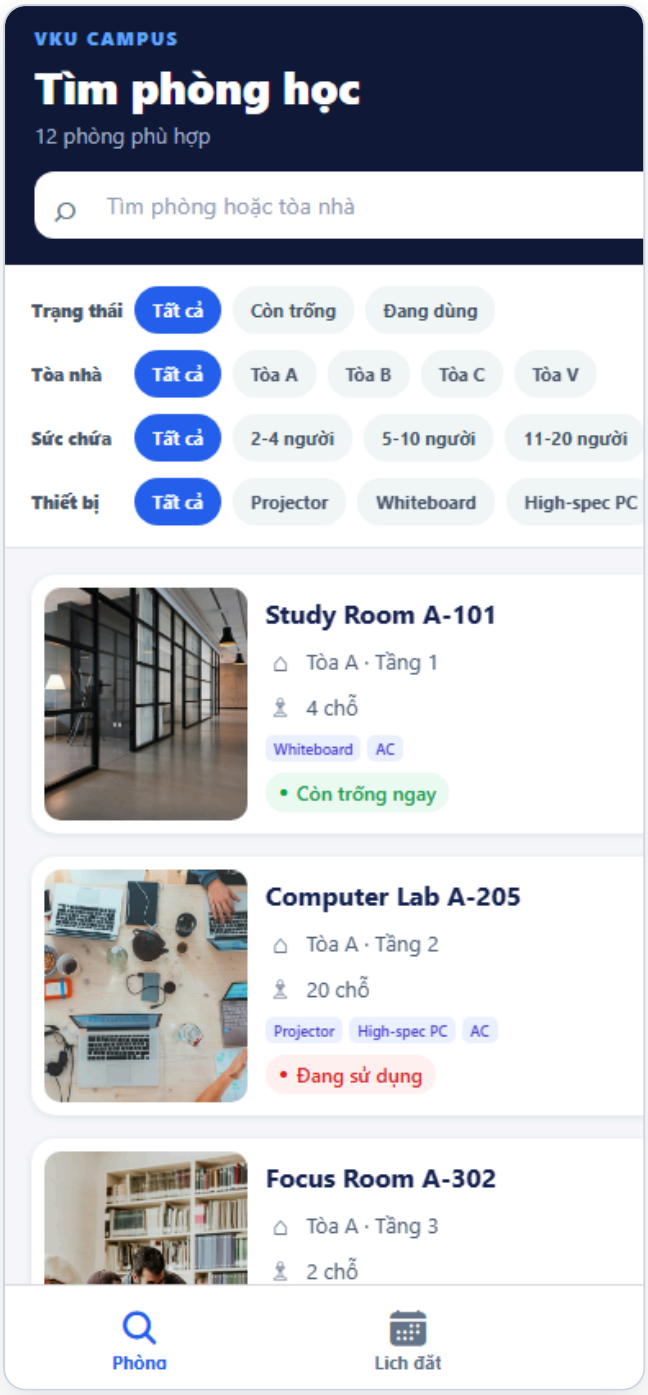
1. **Reliable conflicts:** UI-only disabling can become stale. The interval algorithm is centralized in a service and enforced again at the state boundary.
2. **Cross-platform notifications:** native response APIs are not exposed on web. Platform guards preserve the browser demo, while Android/iOS use local date triggers and deep-link into the booking pass.
3. **Offline continuity:** Zustand's persistence middleware serializes the minimum booking/session slice to AsyncStorage; denied notifications never invalidate a reservation.

### Technology

Expo SDK 57 · React Native 0.86 · TypeScript · Expo Router · Zustand · AsyncStorage · React Query · Reanimated · Expo Notifications · SVG QR

EMPIRICAL EVIDENCE

# 4. Application screenshots




**A. Room discovery.** Search and four combinable filter rows sit above the virtualized list. Every card shows building, capacity, equipment, and current availability.

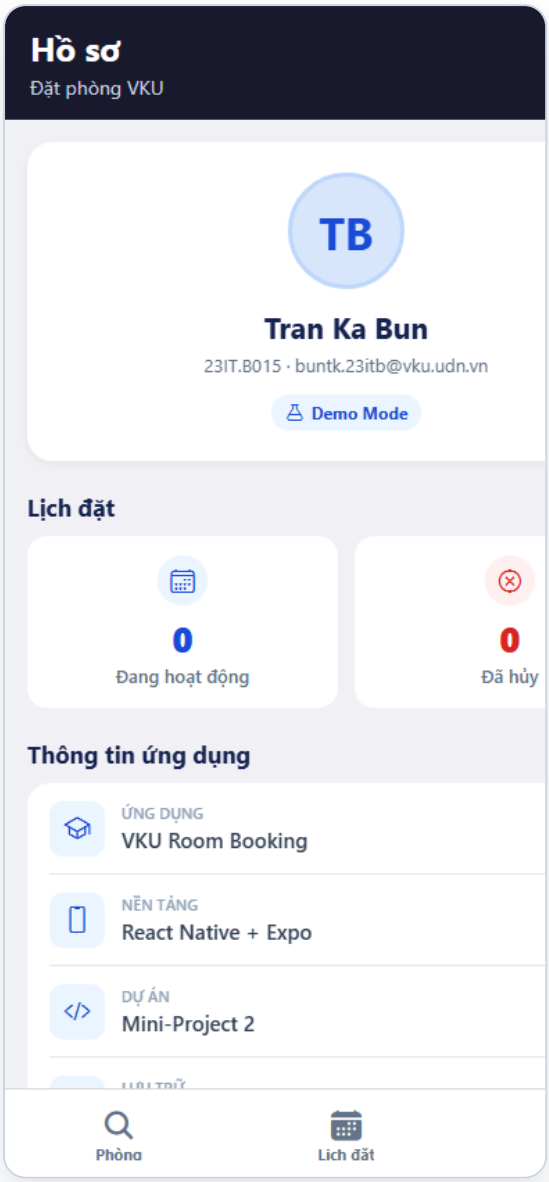



**B. Reservation selector.** Seven-day navigation and fixed two-hour slots. The action remains disabled until a valid slot is selected; occupied slots cannot be pressed.

**Viewport:** Both screenshots were captured from the running Expo web build at 390 × 844 pixels, matching a modern mobile viewport.

EVIDENCE & VERIFICATION

# Persisted identity and quality checks




## C. Student profile

The app stores and displays the required identity:

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Active and cancelled booking totals are derived directly from the persisted store.

## Verification results

- ✓ `npm run lint` — PASS
- ✓ `npm run typecheck` — PASS
- ✓ `npx expo-doctor` — 21/21 PASS
- ✓ `npx expo export --platform web` — PASS
- ✓ Browser smoke test — PASS

## Acceptance summary

Room discovery, filters, scheduling, conflict prevention, offline persistence, QR check-in, cancellation and local reminders are implemented. The app builds for Expo and the responsive web target.

## Known limitation

There is no authenticated backend or multi-device synchronization. A demo video remains a delivery item.